

Why is translation energy a continuum and not quantized?

Asked 7 years, 3 months ago Modified 7 years, 3 months ago Viewed 2k times

▲ When starting the study of quantum theory, usually the teacher separates the functions in electronic, vibrational, rotational and translational parts. As far as I know, we should solve Schrodinger equation for each movement; and as far as I know, rotational and vibrational energies are quantized.

2 ▼ Is there any good explanation about why translational energy is not quantized? And how do we know ψ is separable into vibrational, translational and so on?

🔖 [quantum-mechanics](#) [energy](#) [schrodinger-equation](#) [discrete](#)



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edited Sep 14, 2017 at 5:30

asked Sep 13, 2017 at 23:42

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1 Answer

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▲ If you are in a finite system, translational energy is quantized as well.

2 ▼ The simple example is the infinite square well potential, which is just a finite one-dimensional free particle (also often called a particle in a box). The fact that the wave function must go to zero on the boundary means that the energy levels are sinusoids with wavelengths that evenly divide into $2L$ where L is the size of the system.

🔖 This gives a series of energy levels $E_n = \frac{1}{2m} \frac{\hbar^2}{\lambda_n^2} = \frac{\hbar^2}{8mL^2} n^2$ where n is a positive integer, so the energies are quantized. However, notice that the spacing between the energy levels will be proportional to $\frac{1}{L^2}$ so as the system size becomes large, the energy levels are approximately continuously spaced.



🕒 For a rotational system, the energy spectrum will also smooth out as the size gets large (so you don't see any quantum effects on bicycle wheels) but the molecule does not have a macroscopically large extent for its rotational motion (only the size of the molecule) whereas for its translational degree of freedom it has the whole container.

As for vibrational degrees of freedom, a tiny "block on a spring" isn't a lot different in principle from a particle stuck in a box. But here the particle stays confined by the spring microscopically close to home, so it's effectively a small box and so there is quantization.

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answered Sep 14, 2017 at 0:52

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